

LOW STYRENE SBR PATENT LANDSCAPE: SOLUTION-PHASE STYRENE-BUTADIENE RUBBER WITH TIN-MODIFIED CONTINUOUS POLYMERIZATION

1. ABSTRACT

This report synthesizes the patent evidence for low-styrene styrene-butadiene rubber (SBR) with emphasis on polymerization route, monomer feed, and process control. The selected patent landscape in the provided manifest is narrow and centers on a single primary patent, US6313232B1, which discloses continuous solution polymerization of a conjugated diene monomer or a diene/vinyl aromatic copolymer in hydrocarbon solvent using an organolithium initiator. Although the patent is not explicitly framed as a low-styrene invention, it is directly relevant because it covers butadiene/styrene copolymerization, the core chemistry used to make SBR, and it provides process conditions that influence copolymer architecture and downstream tire performance.

The disclosed synthesis approach is a continuous hydrocarbon-solution process with staged tin-compound addition during polymerization and again after withdrawal of the polymerization mixture. The patent emphasizes that adding tin compounds at 25–80% conversion, preferably 50–80%, and then again inline after reactor discharge improves hysteresis-related properties, including reduced Payne effect and lower tan delta at 50 °C. The evidence also identifies the use of organolithium initiators, hydrocarbon solvents such as hexane, and tin compounds including tin tetrachloride, dibutyltin dichloride, and tributyltin chloride. The examples show post-reaction termination with isopropanol, solvent removal by drum drying, and compounding with carbon black and conventional tire additives.

From a low-styrene SBR perspective, the patent is most useful as a process-control reference rather than a composition-specific low-styrene disclosure. It demonstrates how continuous solution polymerization and post-polymerization tin treatment can be used to tailor rubber performance in butadiene/styrene systems, which is relevant to industrial SBR grades where styrene content is adjusted to meet rolling resistance and wet traction targets.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US6313232B1
- Patent Title: Continuous addition of tin halides to solution SBR
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2001
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses continuous solution polymerization of a conjugated diene monomer or a mixture of a conjugated diene monomer and a vinyl aromatic hydrocarbon monomer in a hydrocarbon solvent using an organolithium initiator. A tin compound is added during polymerization at a specified conversion window and then added again after the polymerization mixture is withdrawn from the reactor, producing diene polymers with improved hysteresis-related properties. - Conversion at first tin addition: between 25% and 80% — Claims / synthesis passages - Preferred conversion at first tin addition: between 50% and 80% — Claims / synthesis passages - Polymerization medium: hydrocarbon solvent — Abstract / claims - Initiator: organolithium compound — Abstract / claims - Tin compound examples: tin tetrachloride, dibutyltin dichloride, tributyltin chloride — Claim 3 / synthesis passages - Polymer cement withdrawal: approximately 300-400 g portions — Comparative Example - Tin compound solution concentration: 10% by weight in hexane — Comparative Example - Tin addition amount: equivalent to 0.04 g active lithium per 100 g monomer — Comparative Example - Mixing time after tin addition: 15-30 minutes — Comparative Example - Compounding temperature: 90 °C — Comparative Example - Compounding temperature: 105 °C — Comparative Example - Dynastat test temperature: 50 °C — Comparative Example - Strain sweep test temperature: 25 °C — Comparative Example - Strain sweep start strain: 0.25% — Comparative Example - Strain sweep increment: 0.25% strain — Comparative Example - Strain sweep endpoint: 14.7% strain — Comparative Example - Sample dimensions: 9.3 mm diameter × 15.5 mm high — Comparative Example - Polymer molecular weight: greater than about 100,000 — synthesis passages - Medium & Water Role: Hydrocarbon-solution polymerization is the core reaction medium. No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - Hydrocarbon-solution polymerization is the core reaction medium. No water-related process step disclosed in the extracted evidence. - Core reaction medium: hydrocarbon solvent solution polymerization

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Example: Post-withdrawal addition of Bu₃SnCl or SnCl₄ to polymer cement improved Dynastat and Payne-effect results versus control. - Comparative Example: Drum drying removed solvents before compounding. - Comparative Example: Repetition with other tin compounds reportedly gave substantially similar results.

Technical Relevance This patent is relevant because it directly covers solution-polymerized butadiene/styrene copolymers, the standard SBR platform, and shows how staged tin treatment can improve tire-relevant properties. It does not explicitly define a low-styrene grade, but it provides a process route that can be applied to SBR compositions with reduced styrene content.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content below conventional SBR grades	Key Findings
US6313232B1	Bridgestone Corporation	2001	Hydrocarbon-solution polymerization is the core reaction medium. No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Comparative Example: Post-withdrawal addition of Bu ₃ SnCl or SnCl ₄ to polymer cement improved Dynastat and Payne-effect r

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Fewer than two selected primary patents were available, so cross-patent quantitative trends were not generated.

4. CONCLUSION

The manifest contains one primary patent, US6313232B1, which is a solution-polymerization process patent for butadiene/styrene copolymers rather than an explicit low-styrene composition patent. Its main contribution to the low-styrene SBR topic is process control: continuous hydrocarbon-solution polymerization with organolithium initiation and staged tin-compound addition to improve hysteresis, Payne effect, and rolling resistance in SBR-type elastomers.

5. REFERENCES

- US6313232B1 | Continuous addition of tin halides to solution SBR | Bridgestone Corporation | US | 2001 | <https://patents.google.com/patent/US6313232B1/en>